

MODULE1

TUTORIAL 2

Roll no

Question

1. In the client-server paradigm, explain why a server should be run all the time, but a client can be run whenever it is needed
2. Can a program written to use the services of UDP be run on a computer that has installed TCP as the only transport-layer protocol? Explain.
3. During the weekend, Alice often needs to access files stored on her office desktop, from her home laptop. Last week, she installed a copy of the FTP server process on her desktop at her office and a copy of the FTP client process on her laptop at home. She was disappointed when she could not access her files during the weekend. What could have gone wrong?
4. Most of the operating systems installed on personal computers come with several client processes, but normally no server processes. Explain the reason.
5. Assume that we add a new protocol to the application layer. What changes do we need to make to other layers?
6. Explain which entity provides service and which one receives service in the client-server paradigm.
7. A source socket address is a combination of an IP address and a port number. Explain what each section identifies.
8. Explain how a client process finds the IP address and the port number to be inserted in a remote socket address.
9. A new application is to be designed using the client-server paradigm. If only small messages need to be exchanged between the client and the server without the concern for message loss or corruption, what transport-layer protocol do you recommend?
10. Alice has a video clip that Bob is interested in getting; Bob has another video clip that Alice is interested in getting. Bob creates a web page and runs an HTTP server. How can Alice get Bob's clip? How can Bob get Alice's clip?
11. When an HTTP server receives a request message from an HTTP client, how does the server know when all headers have arrived and the body of the message is to follow?
12. If an HTTP request needs to run a program at the server site and downloads the result to the client server, the program is an example of
 - a. a static document
 - b. a dynamic document
 - c. an active document

13. Assume that we design a new client-server application program that requires persistent connection. Can we use UDP as the underlying transport-layer protocol for this new application?
14. In FTP, which entity (client or server) starts (actively opens) the control connection? Which entity starts (actively opens) the data transfer connection?
15. What do you think would happen if the control connection were severed before the end of an FTP session? Would it affect the data connection?
16. In FTP, if the client needs to retrieve one file from the server site and store one file on the server site, how many control connections and how many data-transfer connections are needed?
17. In FTP, can a server retrieve a file from the client site?
18. In FTP, can a server get the list of the files or directories from the client?
19. FTP can transfer files between two hosts using different operating systems with different file formats. What is the reason?
20. In a nonpersistent HTTP connection, how can HTTP inform the TCP protocol that the end of the message has been reached?
21. Can you find an analogy in our daily life as to when we use two separate connections in communication similar to the control and data connections in FTP?
22. FTP uses two separate well-known port numbers for control and data connection. Does this mean that two separate TCP connections are created for exchanging control information and data?
23. FTP uses the services of TCP for exchanging control information and data transfer. Could FTP have used the services of UDP for either of these two connections? Explain.
24. Can we have a control connection without a data-transfer connection in FTP? Explain.
25. Can we have a data-transfer connection without a control connection in FTP? Explain.
26. Assume that we need to download an audio using FTP. What file type should we specify in our command?
27. Both HTTP and FTP can retrieve a file from a server. Which protocol should we use to download a file?
28. Does FTP have a message format for exchanging commands and responses during control connection?
29. Does FTP have a message format for exchanging files or a list of directories/ files during the file-transfer connection?
30. Alice has been on a long trip without checking her e-mail. She then finds out that she has lost some e-mails or attachments her friends claim they have sent to her. What can be the problem?
31. Are the HELO and MAIL FROM commands both necessary in SMTP? Why or why not?
32. In Figure 2.20 in the text, what is the difference between the MAIL FROM in the envelope and the FROM in the header?

33. In DNS, which of the following are FQDNs and which are PQDNs?
xxx b. xxx.yyy.net c. zzz.yyy.xxx.edu.
34. Draw a diagram to show the use of a proxy server that is part of the client network:
- Show the transactions between the client, proxy server, and the target server when the response is stored in the proxy server.
 - Show the transactions between the client, proxy server, and the target server when the response is not stored in the proxy server
35. In Chapter 1, we mentioned that the TCP/IP suite, unlike the OSI model, has no presentation layer. But an application-layer protocol can include some of the features defined in this layer if needed. Does HTTP have any presentation layer features?
36. HTTP version 1.1 defines the persistent connection as the default connection. Using RFC 2616, find out how a client or server can change this default situation to nonpersistent
37. In Chapter 1, we mentioned that the TCP/IP suite, unlike the OSI model, has no session layer. But an application-layer protocol can include some of the features defined in this layer, if needed. Does HTTP have any session-layer features?
38. Assume that there is a server with the domain name www.common.com.
- Show an HTTP request that needs to retrieve the document /usr/users/doc. The client accepts MIME version 1, GIF or JPEG images, but the document should not be more than four days old.
 - Show the HTTP response to part a for a successful request.
39. In HTTP, draw a figure to show the application of cookies in a scenario in which the server allows only the registered customer to access the server.
40. In HTTP, draw a figure to show the application of cookies in a web portal using two sites.
41. In HTTP, draw a figure to show the application of cookies in a scenario in which the server uses cookies for advertisement. Use only three sites.
42. In SMTP, a sender sends unformatted text. Show the MIME header
43. POP3 protocol has some optional commands (that a client/server can implement). Using the information in RFC 1939, find the meaning and the use of the following optional commands:
a. UIDL b. TOP 1 15 c. USER d. PASS
44. In FTP, a user (Jane) wants to retrieve an EBCDIC file named huge from /usr/users/report directory using the ephemeral port 61017. The file is so large that the user wants to compress it before it is transferred. Show all the commands and responses.
45. In FTP, a user (Jan) wants to make a new directory called Jan under the directory /usr/users/letters. Show all of the commands and responses.
46. In FTP, a user (Maria) wants to move a file named file1 from /usr/users/report directory to the directory /usr/top/letters. Note that this is a case of renaming a file. We first need to give the name of the old file and then define the new name. Show all of the commands and responses.

47. In Chapter 1, we mentioned that the TCP/IP suite, unlike the OSI model, has no presentation layer. But an application-layer protocol can include some of the features defined in this layer if needed. Does FTP have any presentation-layer features?
48. In Chapter 1, we mentioned that the TCP/IP suite, unlike the OSI model, has no presentation layer. But an application-layer protocol can include some of the features defined in this layer, if needed. Does SMTP have any presentation-layer features?
49. In Chapter 1, we mentioned that the TCP/IP suite, unlike the OSI model, has no session layer. But an application-layer protocol can include some of the features defined in this layer, if needed. Does SMTP or POP3 have any session layer features?
50. In FTP, assume a client with user name John needs to store a video clip called video2 on the directory /top/videos/general on the server. Show the commands and responses exchanged between the client and the server if the client chooses ephemeral port number 56002.
51. Explain the working of BitTorrent with a neat diagram. Discuss the roles of: Torrent file, Tracker, Swarm, Seed, Leech, Choking, Optimistic Unchoking, DHT
52. Compare the following P2P architectures: Centralized P2P (Napster), Unstructured P2P (Gnutella), Structured P2P (BitTorrent). Discuss their advantages and disadvantages.
53. Explain the Domain Name System (DNS). Why is DNS implemented as a distributed system instead of a centralized one?
54. Explain the architecture of an E-mail system. Discuss the roles of User Agent (UA), Message Transfer Agent (MTA), and Message Access Agent (MAA).
55. A user wants to download a binary file named project.zip from the directory /usr/files using FTP. Write the sequence of FTP commands and responses.
56. Explain the architecture of FTP. Why does FTP use two TCP connections instead of one?
57. What are Cookies? Explain how cookies help maintain user sessions.
58. What is a URL? Explain its components using a suitable example
59. Explain the complete BitTorrent protocol with neat diagrams and peer interactions.
60. A client enters the URL www.vjcet.org. Explain how DNS resolves the name into an IP address before communication begins.
61. Explain how BitTorrent downloads a 10 GB file more efficiently than a traditional client-server system when many users are downloading simultaneously.